Multiscale Flow-Based Anomaly Detection for Early Alzheimer's Disease

Amine Agountaf
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Abstract

Early detection of Alzheimer's disease remains a major clinical challenge due to subtle structural brain changes and limited labeled pathological data. This work proposes an unsupervised deep learning framework for Alzheimer's detection using multiscale feature extraction and flow-based anomaly modeling. The model learns the distribution of healthy brain anatomy from MRI images and highlights abnormal regions as deviations from this distribution. By combining convolutional neural networks, spatial attention, cross-scale feature fusion, and invertible normalizing flows, the proposed method produces interpretable anomaly maps that may support early screening and clinical decision-making.

1 Introduction

Alzheimer's disease *(AD)*, the most common form of dementia, is a progressive neurodegenerative disorder characterized by irreversible cognitive decline. Or a complex disease characterized by an accumulation of β-amyloid ($A\beta$) plaques and neurofibrillary tangles composed of tau amyloid fibrils associated with synapse loss and neurodegeneration leading to memory impairment and other cognitive problems. There is currently no known treatment that slows the progression of this disorder[4].

Structural brain changes such as hippocampal atrophy and cortical thinning appear years before clinical diagnosis. However, early detection remains difficult due to subtle imaging patterns and the scarcity of labeled medical data. This project explores an unsupervised anomaly detection approach that models healthy brain anatomy and detects deviations indicative of neurodegeneration.

2 Related Work

Previous approaches to Alzheimer's detection have largely relied on supervised convolutional neural networks trained on labeled MRI datasets. Autoencoders and variational autoencoders have been explored for anomaly detection, but often suffer from blurry reconstructions and weak density estimation. More recently, flow-based models have shown promise due to their exact likelihood estimation and invertible transformations. This work builds upon these ideas by combining multiscale CNN features with flow-based anomaly detection.

3 Dataset and Preprocessing

The model is trained using brain MRI scans from healthy control subjects. Only normal samples are used during training (even if they were less that 700 samples) to learn the distribution of healthy anatomy. This is the proportion of our dataset.

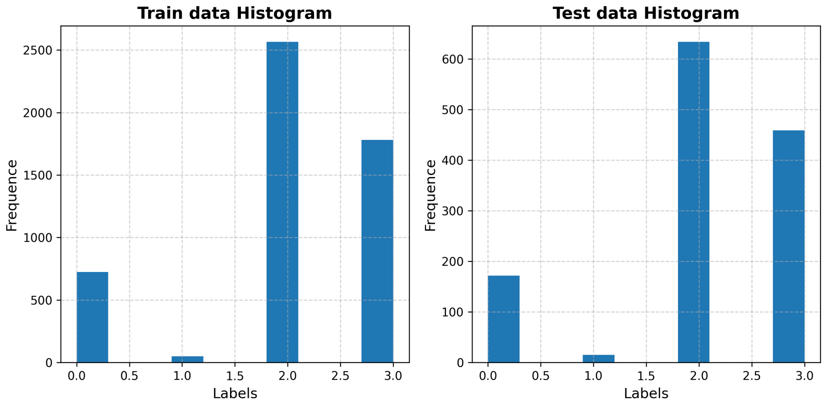

Figure 1: Train and test data histograms

Preprocessing steps include intensity normalization, slice selection, and spatial resizing. This strategy avoids reliance on labeled Alzheimer's data and improves generalization for future improvements. For now only dataset provided by Hack4Health.

4 Methodology

4.1 Model Architecture Overview

The proposed architecture consists of three main components:

- A multiscale convolutional feature extractor
- Cross-scale feature fusion with spatial attention
- Flow-based anomaly detection modules

4.2 Multiscale Feature Extraction

A ResNet-based backbone extracts hierarchical feature maps at multiple spatial resolutions. Spatial attention mechanisms emphasize anatomically relevant regions such as the hippocampus and cortical areas.

4.3 Cross-Scale Feature Fusion

To combine global and local anatomical information, higher-level features are projected and fused into lower-level representations. This allows fine-grained texture features to benefit from global contextual information.

4.4 Flow-Based Anomaly Modeling

Each scale is modeled using a FastFlow-style normalizing flow composed of invertible 1×1 convolutions and affine coupling layers. The flow learns the probability density of healthy features and assigns low likelihood to anomalous regions.

4.5 Anomaly Map Generation

Anomaly maps are generated by computing the energy of latent variables and aggregating results across scales. The final output is a pixel-wise heatmap highlighting regions that deviate from healthy anatomy.

5 Results

The model is trained in an unsupervised manner using only healthy samples. Optimization is performed by minimizing the negative log-likelihood of the flow models as in this figure 2. This approach reduces overfitting and removes the need for explicit pathological labels.

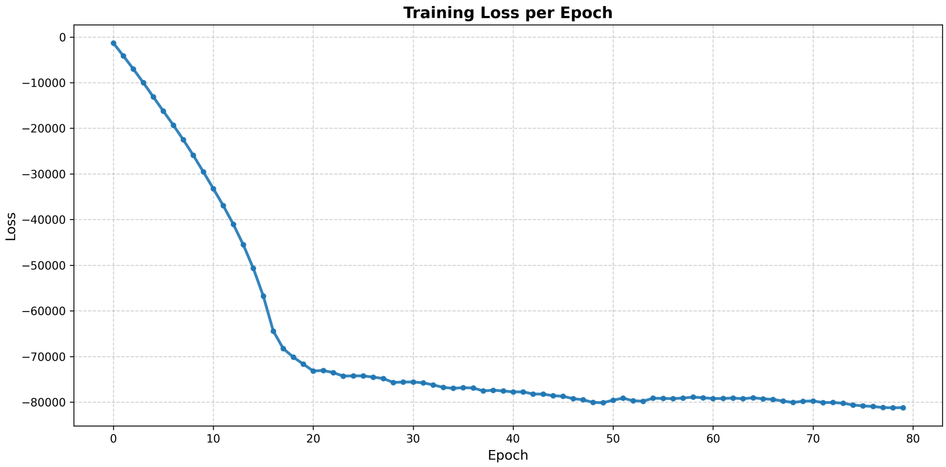

Figure 2: Loss function

The proposed method produces coherent anomaly maps that align with known Alzheimer-related brain regions. Qualitative evaluation shows increased sensitivity in hippocampal and cortical areas, while maintaining robustness to scanner variability. These results suggest the model's potential for early-stage detection as we see in figure 3.

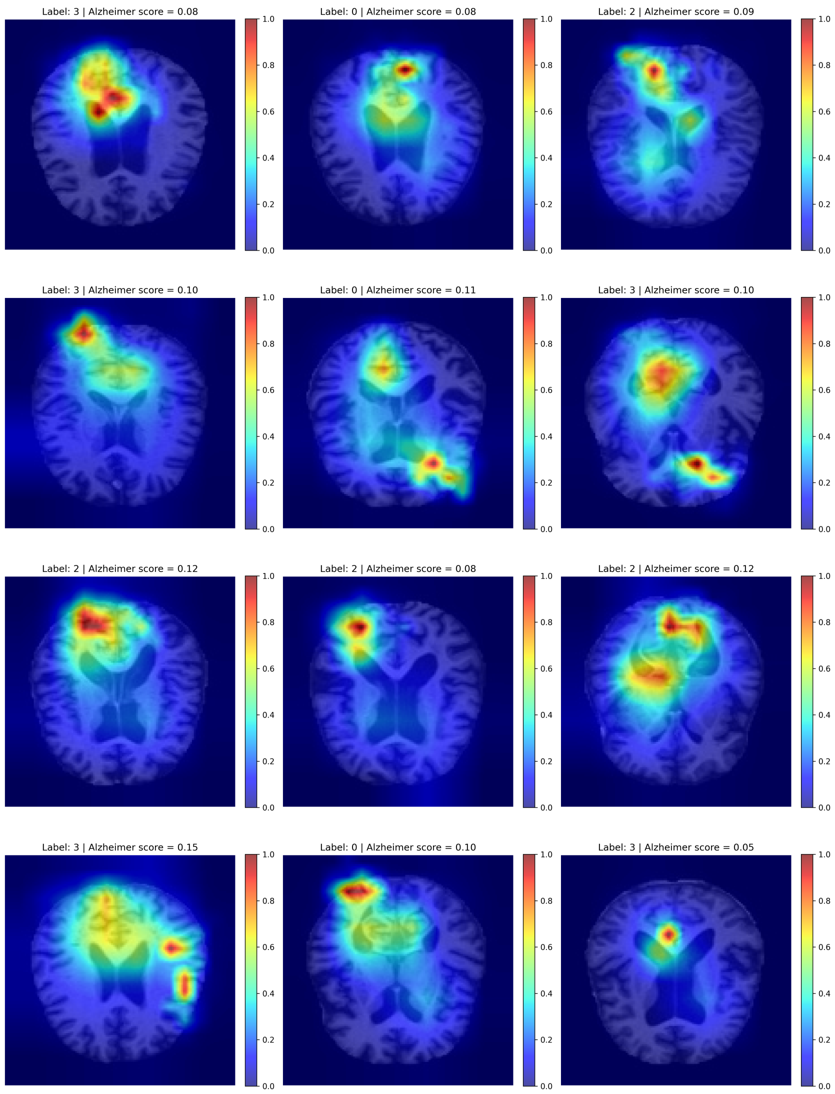

Figure 3: Anomaly map for 12 test MRI samples.

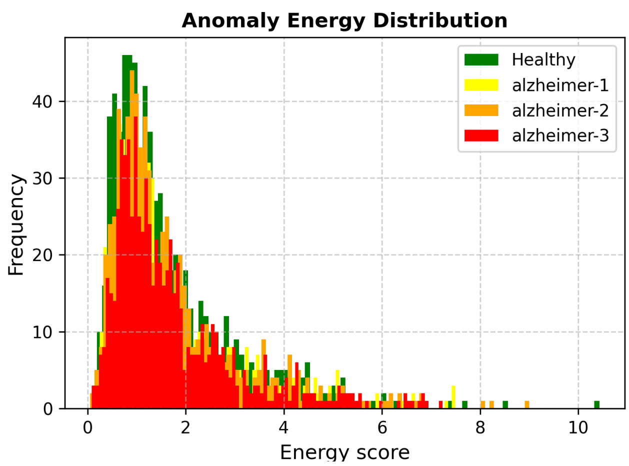

Figure 4: Energy distributions for each label in test dataset.

The anomaly energy distributions show substantial overlap between healthy and early Alzheimer's subjects, reflecting the subtle and localized nature of neurodegeneration. Importantly, a progressive right-shift and heavier tails are observed with increasing disease severity, indicating that the proposed model captures stage-dependent structural deviations rather than trivial global differences.

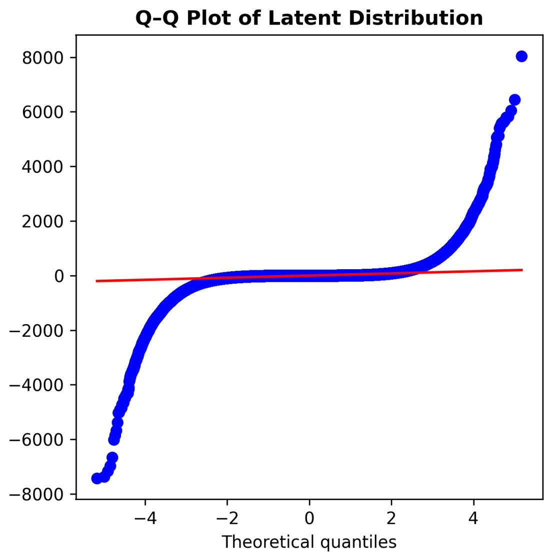

Figure 5: Quantile-Quantile Latent Distribution

This Q–Q plot stands for Quantile–Quantile latent distribution, the quantiles of your data against the quantiles of a reference distribution which is Normal *(or Gausssian)* distribution. Here we see strong deviation in both tails, heavy-tailed behavior and central region roughly linear. This Deviations from the diagonal line, particularly in the tails, indicate non-Gaussian behavior associated with rare or pathological patterns. This behavior is expected and desirable in anomaly detection, as abnormal samples are intentionally mapped to low-probability regions of the latent space.
In summary: **The Q–Q plot verifies how closely the learned latent space follows the Gaussian prior and highlights deviations caused by pathological brain structures.**

6 Conclusion

This project demonstrates the feasibility of using multiscale flow-based anomaly detection for early Alzheimer's disease screening. The primary strengths of this approach include unsupervised learning, multiscale reasoning, and interpretability. By learning normal brain anatomy and highlighting deviations, the proposed framework offers an interpretable and scalable approach for supporting early diagnosis. In the other hand, we see some limitations including the use of 2D slices instead of full 3D volumes and the need for further clinical validation. This system is intended as a decision-support tool and not as a standalone diagnostic system. Clinical deployment would require extensive validation, bias analysis, and compliance with patient privacy regulations. Future improvements may address these limitations which can include extending the model to 3D volumetric MRI, longitudinal disease progression analysis, and integration with cognitive and clinical biomarkers.

References

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